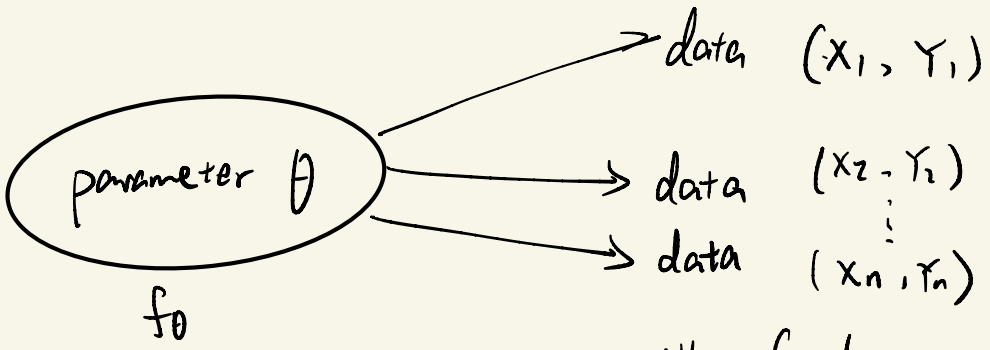
Week 1 Gaussian Sequence Model

(see also typed lecture notes)



$$Y_i = f_\theta(X_i) + \text{noise}$$

Y_i: responses; X_i: predictors; X_i: either fixed or random

Linear regression: f_θ is linear function

$$f_\theta(x_i) = \theta^T x_i, \qquad x_i \in \mathbb{R}^d, \ \theta \in \mathbb{R}^d$$

$$Y = X\theta + \text{noise}$$

$$\begin{bmatrix} \\ \\ \end{bmatrix} = \begin{bmatrix} X_1^T \\ \vdots \\ X_n^T \end{bmatrix} \theta + \text{noise}$$

$$\text{noise} = \begin{pmatrix} \xi_1 \\ \vdots \\ \xi_n \end{pmatrix}$$

Gaussian linear model: $\xi_1 \cdots \xi_n$ iid $N(0, \sigma^2)$

- Fixed design: $X = \begin{bmatrix} X_1^T \\ \vdots \\ X_n^T \end{bmatrix}$ is deterministic
- Random design: $X_1 \cdots X_n$ iid random vectors in $\mathbb{R}^d$.

Gaussian sequence model: $X = Id$

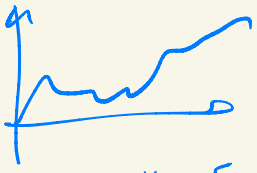
$$\Leftrightarrow \quad Y_i = x_i + \xi_i$$

Remark: in nonparametric regression,

$$S(x) = f(x) + \frac{dW(x)}{dx}$$

$$\Leftrightarrow$$

$$S_i = F_i + \xi_i$$



W: Brownian motion $[0,1]$

S: observation

f: unknown smooth function

F: Fourier trans. of f

S: Fourier of s

$\{\xi_i\}$: Fourier of $\{\frac{dw}{dx}\}$

Define: $\|\theta\|_0 = \sum_{i=1}^{d} 1_{\theta_i \neq 0}$

$$B_0(k) = \{\theta : \|\theta\|_0 = k\}.$$

naive: $\hat{\theta}(Y) = Y$

$$\text{Risk} = \mathbb{E}\|\hat{\theta} - \theta\|_2^2 = \mathbb{E}\|\xi\|_2^2 = d\sigma^2$$

oracle: if $\text{supp}(\theta)$ is known:

$$\hat{\theta}_i^{\text{oracle}} = Y_i 1_{i \in \text{supp}(\theta)}$$

$\Rightarrow$ Risk $= k\sigma^2$

Improves naive estimator drastically if $k \ll d$.

Hard thresholding estimator:

Input: $\theta \in \mathbb{R}^d$, $\tau \in (0, \infty)$

Output: $\hat{\theta}$

$$\hat{\theta}_i = Y_i \mathbb{1}_{|Y_i| > \tau}$$

Idea for picking τ: just large enough so that no Type-I error (with constant probability).

$\theta_i = 0$, but $\hat{\theta}_i \neq 0$ (false discovery)

$\Leftrightarrow |Y_i| > \tau$ for HT

want: $\mathbb{P}[\exists i \le d : |\xi_i| > \tau_0] < \delta$

By union bound,

$$\mathbb{P}[\dots] \le d \cdot \mathbb{P}[|\xi_i| > \tau_0]$$

$$\le d \cdot 2 \cdot \mathbb{P}[\xi_i > \tau_0]$$

$$\le 2d\, e^{-\tau_0^2/2\sigma^2}$$

want $= \delta$

$$\Rightarrow \quad T_0 = \sigma\sqrt{2\log\frac{2d}{\delta}} \qquad (*)$$

Thm 1: with $T = 2T_0$, where T_0 as in $(*)$,

for any $\theta \in B_0(k)$, we have:

$$\|\hat{\theta} - \theta\|_2^2 \lesssim \sigma^2 k \log\frac{2d}{\delta}$$

with probability $\geq 1-\delta$

notation: $a \lesssim b$ if $a \leq$ const. b

Example: $k = \sqrt{d} \Rightarrow$

$$\mathbb{E}\|\hat{\theta}^{HT} - \theta\|_2^2 \lesssim \sigma^2\sqrt{d}\log d$$

$$\mathbb{E}\|\hat{\theta}^{ovade} - \theta\|_2^2 \leq \sigma^2\sqrt{d}$$

$$\mathbb{E}\|\hat{\theta}^{naive} - \theta\|_2^2 \leq \sigma^2 d$$

Proof: with probability $1-\delta$,

$$\max_{i \leq kd} |\xi_i| \leq T/2 \quad \leftarrow \text{event } \mathcal{A}$$

under $\mathcal{A}$,

$$\begin{aligned}
|\hat{\theta}_i - \theta_i| &= |Y_i - \theta_i| 1_{|Y_i| > T} + |\theta_i| 1_{|Y_i| \leq T} \\
&= |\xi_i| 1_{|\theta_i + \xi_i| > T} + |\theta_i| 1_{|Y_i| \leq T} \\
&\leq \frac{T}{2} 1_{|\theta_i + \xi_i| > T} + |\theta_i| 1_{|\theta_i + \xi_i| \leq T} \\
&\leq \frac{T}{2} 1_{|\theta_i| > \frac{T}{2}} + |\theta_i| 1_{|\theta_i| \leq \frac{3}{2}T}
\end{aligned}$$

$$\lesssim \min\{|\theta_i|, \tau\}$$

$$\|\hat{\theta} - \theta\|_2^2 \le \sum_{i=1}^{d} \min\{|\theta_i|^2, \tau^2\} \lesssim \|\theta\|_0 \tau^2$$

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Remark(1): HT is actually adaptive !

b/c $\tau = 2.2\sigma\sqrt{2\ln\frac{2d}{\delta}}$ which doesn't depend on k.

(2): For soft-thresholding estimator where

$$\hat{\theta}_i^{soft} := (Y_i - \tau)1_{Y_i > \tau} + (Y_i + \tau)1_{Y_i < -\tau}$$

we have similar properties (i.e. $Risk = O(\sigma^2 k \ln\frac{2d}{\delta})$ w.p. $1-\delta$)

